

Empirical Evaluation of the Framework for Truth-Probability, Perceived Credibility, and Importance Scoring

A Three-Dataset External-Scoring Study Using Blind LLM-Based Variable Assignment

Empirical Study Paper — V3.5

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Abstract

This empirical study evaluates a unified framework for scoring statements along three distinct outputs: evidence-weighted truth-probability $T_{abs}(s)$, receiver-dependent perceived truth-probability $T_{perc}(s,r)$, and receiver-dependent importance $I(s,r)$. The study uses a blind external-scoring design in which GPT-5.5 Thinking assigns atomic framework variables rather than final benchmark labels. Final scores are then computed by the mathematical formula engine and joined to hidden targets only after scoring is complete. Three datasets are used because each corresponds to a different output type: HumAID for crisis-monitoring importance, CREDBANK/CredEvent TurkRatings for perceived credibility, and LIAR/PolitiFact for evidence-weighted truth. Across the three tests, the framework produces moderate-to-strong empirical alignment with benchmark targets. The strongest evidence is in discriminative and ranking performance, with binary AUC values in the acceptable-to-excellent range and correlations reaching large-effect territory on the harder credibility and factuality tasks. The results support the framework as a structured scoring architecture while identifying calibration issues that remain most visible at distributional extremes, especially in graded factuality scoring.

Keywords: truth-probability; perceived credibility; importance scoring; external scoring; blind LLM scoring; LIAR; HumAID; CREDBANK; AUC; calibration

1. Introduction

The theoretical framework evaluated here assigns three scores to a statement s made by a source to a receiver r : $T_{\text{abs}}(s)$, an evidence-weighted truth-probability; $T_{\text{perc}}(s,r)$, a perceived truth-probability; and $I(s,r)$, a receiver-dependent importance score. The empirical question is whether this structured, variable-based architecture aligns with external benchmark targets when the final labels are hidden from the scoring agent.

This study is designed as an output-specific validation rather than as a single generic classification task. A crisis tweet may be important without being independently verified; an event stream may appear credible to human raters without being a pure objective truth label; and a political factuality claim requires evidence-sensitive truth assessment rather than contextual importance scoring. For that reason, the three datasets are mapped to different framework outputs.

The scorer is not treated as an all-knowing model. For T_{abs} , the relevant epistemic ideal is a highly unbiased, fairly skeptical expert using available evidence. This distinction matters: evidence-weighted truth-probability is a posterior under an evidence condition, not omniscient access to the world state.

1.1 Contributions

- Tests the framework across three conceptually distinct outputs rather than collapsing all evaluation into a single truth label.
- Uses a target-free external scoring protocol in which the LLM assigns atomic variables and the formula engine computes the final output.
- Evaluates ranking, linear association, calibration error, graded discrimination, and binary discrimination.
- Demonstrates the importance of evidence depth for T_{abs} through a LIAR evidence-research progression, while keeping this as a secondary diagnostic rather than the paper's central focus.
- Provides a reproducible owner-side analysis pipeline with scorer inputs, external outputs, logs, joined audit files, and computation scripts.

2. Research Questions

Table 1. Research questions and output mapping.

RQ	Question	Linked output
RQ1	Can the framework align with humanitarian importance/relevance judgments in crisis communication data?	$I(s,r)$
RQ2	Can the framework align with human perceived-credibility judgments for event-related social-media claims?	$T_perc(s,r)$
RQ3	Can the framework align with graded factuality labels when claims are evaluated using evidence-based external scoring?	$T_abs(s)$
RQ4	For evidence-weighted truth, does more extensive non-verdict evidence research improve alignment with graded factuality labels?	$T_abs(s)$ diagnostic

RQ4 is included because evidence depth is especially central to T_abs . It is treated as a methodological diagnostic rather than as the main empirical claim. The primary empirical structure remains the three-output mapping: importance, perceived credibility, and evidence-weighted truth.

3. Framework Outputs and Empirical Mapping

The framework separates evidence-weighted truth, perceived truth, and importance. This separation is essential to the empirical design because the benchmark target attached to each dataset is not the same type of quantity.

Table 2. Framework outputs and empirical correspondences.

Output	Meaning in the framework	Primary input clusters	Dataset used
$T_abs(s)$	Evidence-weighted truth-probability assigned by a highly unbiased skeptical expert using available evidence.	source credibility; statement properties; corroboration/evidence	LIAR / PolitiFact
$T_perc(s,r)$	Receiver-dependent perceived truth-probability or credibility.	T_abs plus delivery, situational context, and receiver properties	CREDBANK / CredEvent TurkRatings
$I(s,r)$	Receiver-dependent importance or consequence of the statement.	delivery, relevance-gated situation, receiver consequence, dampened truth update	HumAID

The relevance gate $Rel(s,c)$ is not the identity of the whole framework; it is one important mechanism inside the situational-context path. Its role is to prevent ambient context from inflating importance unless the statement itself bears on the relevant stakes. The empirical study therefore refers to the framework as a truth-probability, perceived-credibility, and importance-scoring framework, not merely as a relevance-gated model.

4. Datasets

Table 3. Dataset selection and target interpretation.

Dataset	Source domain	Sample used	Target type used in this study	Reason for use
HumAID	Human-annotated disaster tweets	200 blind rows	Humanitarian importance / relevance mapped to $I(s,r)$	Crisis-monitoring relevance is naturally receiver-dependent and consequence-sensitive.

CREDBANK / CredEvent TurkRatings	Twitter event credibility judgments	200 blind rows	Human-perceived credibility mapped to $T_perc(s,r)$	Crowd credibility ratings are closer to perceived credibility than to omniscient truth.
LIAR / PolitiFact	Short political factuality claims	200 blind rows	Graded factuality mapped to $T_abs(s)$	Political truth claims require evidence-sensitive skeptical-expert assessment.

HumAID is a human-annotated disaster tweet dataset developed for crisis informatics and humanitarian information classification. CREDBANK/CredEvent provides social-media event credibility judgments from human annotators. LIAR contains short political statements collected from PolitiFact with six graded truthfulness labels. These sources differ in object of evaluation, which makes them well suited to testing the framework’s output separation.

5. External Blind Scoring Protocol

The external scorer used for variable assignment was GPT-5.5 Thinking. The model did not receive benchmark target values and was not asked to predict final dataset labels directly. Instead, it assigned atomic framework variables under a fixed schema. The final mathematical outputs were computed owner-side using the framework formula engine.

- Scorer inputs were target-free and separated from owner-side target files.
- The scorer received variable definitions, calibration guidance, dataset-specific receiver profiles, and output templates.
- For HumAID and CredEvent, the scorer used blind row context and dataset-specific instructions to assign variables relevant to $I(s,r)$ and $T_perc(s,r)$.
- For LIAR, the scorer was required to conduct extensive outside research for each row while avoiding verdict-bearing fact-check sources and benchmark labels.
- External outputs were returned as CSV files with fixed columns, together with row-level logs for LIAR.
- Targets were joined only after scoring, during owner-side evaluation.

5.1 Leakage control

Leakage control was central to the protocol. For LIAR, the scorer was prohibited from using LIAR labels, PolitiFact verdict or Truth-O-Meter pages, Snopes, FactCheck.org, Washington Post Fact Checker verdict pages, AP/Reuters/AFP fact-check verdict pages, benchmark answer keys, previous model outputs, or target files. The allowed evidence sources included official statistics, government records, roll-call votes, bill histories, archived pages, court records, transcripts, academic/nonpartisan research, and reputable non-verdict news when primary records were unavailable.

6. Formula Implementation

The formula engine implemented the framework-defined computations. T_abs was computed from source credibility, statement properties, and corroboration/evidence variables. T_perc used T_abs as its anchor and updated through delivery, situational, and receiver variables. $I(s,r)$ used the framework’s importance engine with delivery, relevance-gated situational context, receiver consequence, and a dampened truth update. Multiplicative delivery, situational, and receiver composites were root-normalized according to the framework before the importance tilde transformation.

K_count and K_mean were handled as corroboration information. When K_count was zero, no independent corroboration update was applied; when independent sources were present, the evidence direction and

strength were incorporated through K_mean and the evidence variables L and M. N_raw was converted into the framework’s derived novelty terms for perceived truth and importance.

7. Evaluation Metrics and Strength Interpretation

The study reports five complementary metrics. Spearman correlation measures monotonic ranking alignment. Pearson correlation measures linear association. MAE measures absolute calibration error on the 0–1 target scale. Graded AUC, implemented as a concordance index, estimates pairwise ranking discrimination over continuous or graded targets. Binary ROC-AUC is reported where a meaningful threshold is defined.

Metric interpretation is contextual. Cohen-style conventions often treat correlations around 0.10, 0.30, and 0.50 as small, medium, and large effects, respectively, though domain-specific interpretation is recommended. AUC values are commonly read as discrimination above chance, with values in the 0.70–0.80 range often considered acceptable and 0.80–0.90 often considered excellent in applied settings. AUC can also be interpreted as the probability that a randomly chosen positive item is ranked above a randomly chosen negative item.

8. Results

Across all three datasets, the framework produced moderate-to-strong empirical alignment. The results are particularly meaningful because the external scorer did not directly predict target labels; it assigned framework variables, and the final output was produced by the formula engine.

Table 4. Main empirical metrics across the three datasets.

Dataset	Output	n	Spearman	Pearson	MAE	Graded AUC	Binary AUC
HumAID	I(s,r)	200	0.407	0.477	0.158	0.658	0.757
CredEvent	T_perc(s,r)	200	0.591	0.563	0.124	0.719	0.876
LIAR / PolitiFact	T_abs(s)	200	0.630	0.560	0.263	0.764	0.825

Table 5. Mean calibration summary.

Dataset	Model mean	Target mean	Calibration interpretation
HumAID	0.813	0.785	good mean calibration; ranking/discrimination are moderate and useful for triage-style importance
CredEvent	0.882	0.938	strong perceived-credibility alignment with low calibration error
LIAR / PolitiFact	0.436	0.390	strong ranking and discrimination; tail calibration remains the main challenge

8.1 HumAID: Importance I(s,r)

HumAID tests receiver-dependent importance. The framework achieved moderate ranking alignment and acceptable binary discrimination while maintaining good mean calibration. This indicates that the importance engine captures a substantial part of the structure in crisis-monitoring relevance judgments. The strongest remaining diagnostic issue is over-scoring some disaster-adjacent but non-operational content; this is a refinement issue in distinguishing true humanitarian actionability from topic proximity.

Table 6. HumAID group-level diagnostic.

HumAID group	n	Target mean	Model mean	MAE
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requests or urgent needs	20	1.000	0.869	0.131
injured or dead people	20	1.000	0.912	0.088
missing or found people	20	0.950	0.915	0.043
displaced people and evacuations	20	0.950	0.865	0.100
caution and advice	20	0.900	0.892	0.082
infrastructure and utility damage	20	0.900	0.780	0.140
rescue volunteering or donation effort	20	0.800	0.915	0.115
other relevant information	20	0.700	0.751	0.235
sympathy and support	20	0.500	0.590	0.150
not humanitarian	20	0.150	0.637	0.494

8.2 CredEvent: Perceived credibility $T_{perc}(s,r)$

CredEvent provides the strongest and cleanest evidence in the study. Correlations are in the moderate-to-strong range, binary AUC is excellent for distinguishing very credible event claims, and MAE is the lowest of the three datasets. This result is theoretically coherent because the dataset target is a human credibility judgment, which maps naturally to $T_{perc}(s,r)$ rather than to omniscient truth.

Table 7. CredEvent calibration by target bin.

Target bin	n	Target mean	Model mean	MAE
0.4-0.7	1	0.667	0.003	0.664
0.7-0.9	31	0.838	0.589	0.341
>0.9	168	0.958	0.941	0.081

8.3 LIAR / PolitiFact: Evidence-weighted truth $T_{abs}(s)$

LIAR is the strictest test because it evaluates evidence-weighted truth rather than perceived credibility or importance. The V3.5 extensive-research condition produced strong ranking alignment and excellent binary factuality discrimination. Absolute calibration is less strong than ranking discrimination because the model still compresses the extremes: very false claims tend to be scored somewhat too high, and fully true claims somewhat too low. Even so, the ordered label-group pattern is encouraging for a blind variable-assignment protocol.

Table 8. LIAR label-group diagnostic.

LIAR label	n	Target mean	Model mean	MAE	Model-target
pants-fire	34	0.005	0.135	0.134	0.130
false	34	0.100	0.196	0.225	0.096
barely-true	34	0.250	0.344	0.351	0.094
half-true	34	0.400	0.510	0.364	0.110
mostly-true	34	0.700	0.697	0.293	-0.003
true	30	0.950	0.776	0.204	-0.174

8.4 RQ4: Evidence-depth diagnostic for T_{abs}

The LIAR progression shows that T_{abs} performance depends strongly on evidence retrieval depth. A broad-context evidence packet was not enough; requiring outside research produced a large improvement, and requiring extensive exact-claim research improved the result further. This is consistent with the framework's definition of T_{abs} as evidence-weighted skeptical-expert probability: the expert can only update strongly when usable evidence is available.

Table 9. LIAR improvement across evidence-depth conditions.

LIAR condition	Research mode	Spearman	Pearson	MAE	Model mean
V3.3	Broad evidence only	0.069	0.052	0.306	0.395
V3.4	Outside research required	0.525	0.481	0.277	0.477
V3.5	Extensive exact-claim research	0.630	0.560	0.263	0.436

8.5 AUC threshold diagnostics

Table 10. Binary AUC threshold diagnostics.

Dataset	Threshold	AUC	Positives	Negatives
HumAID	target >= 0.5	0.757	180	20
HumAID	target >= 0.75	0.773	140	60
HumAID	target >= 0.9	0.696	120	80
CredEvent	target >= 0.5	—	200	0
CredEvent	target >= 0.9	0.876	169	31
CredEvent	median	0.791	108	92
LIAR / PolitiFact	target >= 0.25	0.823	132	68
LIAR / PolitiFact	target >= 0.5	0.825	64	136
LIAR / PolitiFact	target >= 0.75	0.801	30	170

9. Discussion

The results support the central empirical premise of the framework: different statement-evaluation targets should be mapped to different outputs. CredEvent performs strongly because perceived event credibility matches $T_{perc}(s,r)$. HumAID performs well enough to support $I(s,r)$ as a meaningful consequence-sensitive score, especially at the level of mean calibration and binary relevance discrimination. LIAR demonstrates that T_{abs} can align meaningfully with graded factuality labels when the scorer is allowed and required to resolve exact claims through non-verdict evidence.

The most important methodological result is that the framework is not merely a prompt-based label predictor. The scorer assigns atomic variables, and the formula engine produces the final score. This makes the process more auditable: errors can be traced to specific variables, evidence logs, or calibration behavior rather than being hidden inside a single opaque prediction.

The metrics are decent-to-strong for this setting. Two of the three datasets reach correlations around or above the conventional large-effect range, and binary AUC reaches excellent levels for perceived credibility and factuality discrimination. The HumAID result is more moderate but still useful, especially given the noisy boundary between humanitarian relevance, disaster-adjacent commentary, and non-operational social-media content.

10. Scope Conditions and Limitations

The study has several scope conditions. The sample size is 200 rows per dataset. The external scorer was GPT-5.5 Thinking rather than a panel of independent human raters or multiple model families. LIAR depends on the quality of exact-claim evidence retrieval, and a small number of rows remained unresolved in the final run. Benchmark targets are also imperfect proxies: HumAID labels are mapped to numeric importance, CredEvent is based on human credibility ratings, and LIAR labels are graded fact-checking judgments rather than direct access to truth.

These limitations do not erase the empirical signal. They define the next stage of validation: larger samples, multiple scorers, inter-rater reliability, and additional calibration work at the tails. The present study should be read as a serious external-scoring validation and diagnostic of the framework’s architecture, not as a final endpoint.

11. Conclusion

This three-dataset empirical study provides encouraging support for the framework’s output-specific architecture. HumAID supports the importance path $I(s,r)$ with good calibration and acceptable relevance discrimination. CredEvent gives strong evidence for $T_perc(s,r)$, showing that the framework can recover human perceived credibility judgments. LIAR provides the strongest ranking evidence for $T_abs(s)$ once extensive non-verdict research is required, confirming that evidence depth is central to evidence-weighted truth scoring.

Overall, the framework performs meaningfully across three distinct target types under a blind variable-assignment protocol. The results are strongest for discrimination and ranking and remain promising for calibration, with the clearest remaining work being tail calibration for factuality and finer separation of humanitarian actionability from disaster-adjacent content.

12. Data and Reproducibility Statement

The study was organized into separate target-free scorer-input archives, raw external-output/log archives, and owner-side analysis bundles. The owner-side analysis bundle contains reproducible computation scripts, joined audit CSVs, metrics tables, AUC diagnostics, and provenance notes. Target files were not included in scorer-facing packets. The final reported metrics correspond to the V3.5 analysis package.

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Appendix A. Metric Definitions

Table A1. Metric definitions.

Metric	Definition	Interpretive use in this study
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Spearman	Rank correlation between model score and target score.	Primary measure of monotonic alignment.
Pearson	Linear correlation between model score and target score.	Measures linear association and scale relationship.
MAE	Mean absolute error on the 0-1 target scale.	Primary calibration error measure.
Graded AUC / C-index	Pairwise concordance over continuous or graded targets.	Ranking discrimination for graded labels.
Binary ROC-AUC	Area under ROC curve after thresholding the target.	Thresholded discrimination where meaningful.

Appendix B. Final Packet Versioning

The final scorer-facing package used in this empirical study is V3.5. Earlier iterations documented useful diagnostics: the first LIAR run showed that broad contextual evidence alone was insufficient for T_abs; V3.4 required outside research; V3.5 strengthened the exact-claim research requirement and warned against unwarranted posterior shrinkage toward the middle when direct evidence was strong. These iterations refined the scoring protocol and evidence-access rules, not the theoretical output architecture.